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Fatty acid oxidation has been studied by analyzing the effect of leptin on the activation of AMPK and ACC in soleus muscles. To identify leptin's mode of action, leptin injections were administered to transgenic mice via intravenous (i.v., 1 mg/kg) and intrahypothalamic (i.h.p., 1 ng) routes. The i.v. route delivers leptin systemically to the whole body through the circulatory system. This route reaches muscles quickly but is slow to reach the sympathetic control center, and i.v. leptin is rapidly cleared from the blood (with a half-life of $t_{1/2} = 40.2$ minutes). In contrast, the i.h.p. route delivers leptin directly to the sympathetic nervous system but has no direct action on muscles.

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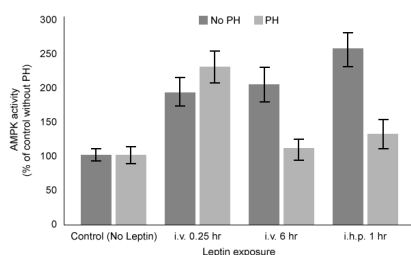
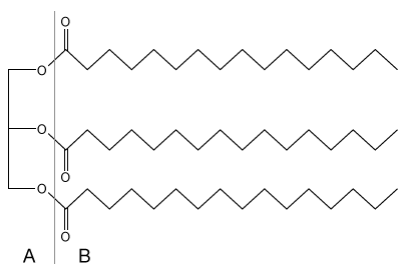


Figure 1 AMPK activity ($\pm$ standard error of the mean) measured in soleus muscle injected with leptin and PH via i.v. and i.h.p.

Table 1 ACC Activity in Soleus Muscle Exposed to Leptin Delivered via i.v. and i.h.p. Injections *in vivo*

Injection	ACC activity ($\text{nmol g}^{-1} \text{min}^{-1}$)	
	0 mM citrate	2 mM citrate
Saline 6 hr i.v.	12	45
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Leptin 1 hr i.h.p.	7	18

The structure of Compound 1, which is found in adipocytes, is shown below. What would be the catabolic fate of this molecule?

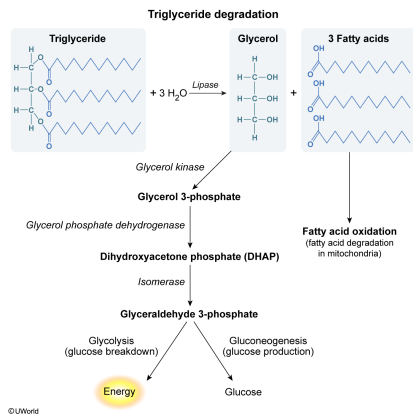


- ✓ ☒ A. Part A enters glycolysis and part B enters β -oxidation. [68%]
- ☐ B. Part A enters gluconeogenesis and part B enters fatty acid synthesis. [12%]
- ☐ C. Part B enters fatty acid synthesis and part A enters β -oxidation. [17%]
- ☐ D. Part B enters glycogenolysis and part A enters glycolysis. [1%]

Correct

69% answered correctly

Explanation:



Compound 1 is a **triglyceride**, which is composed of **three fatty acid chains** (part B) bonded to **glycerol** (part A) by an ester bond (gray line). When lipids must be degraded (ie, catabolized) for fuel, triglycerides stored in adipocytes are first hydrolyzed by hormonally controlled lipases, yielding three fatty acids and one glycerol molecule. The fatty acids are released into the bloodstream and delivered to fuel-dependent tissues by albumin. Subsequently, the **fatty acids** are activated and enter the mitochondria for **β -oxidation**, which is their major catabolic pathway.

Glycerol cannot be metabolized further by adipocytes; it is secreted into circulation and transported to the liver, where it is phosphorylated to glycerol 3-phosphate by glycerol kinase. Glycerol 3-phosphate is then converted to dihydroxyacetone phosphate (DHAP) by glycerol phosphate dehydrogenase. DHAP is isomerized to glyceraldehyde 3-phosphate, an intermediate compound of **glycolysis**.

(Choice B) Although glycerol (part A) can be incorporated into **gluconeogenesis**, it is an anabolic (not a catabolic) pathway. Furthermore, free fatty acids (part B) are the final products of fatty acid synthesis, not the initial reagents.

(Choice C) The reagents for **fatty acid synthesis** are acetyl-CoA and malonyl-CoA. Fatty acid chains with 16 carbon atoms (part B) do not enter fatty acid synthesis. Glycerol (part A) is not involved in fatty acid synthesis but is converted to DHAP for glycolysis.

(Choice D) **Glycogenolysis** is a separate catabolic pathway that breaks down glycogen stored in the liver into glucose to increase ATP production. Glycerol (part A) can enter glycolysis, but free fatty acids (part B) are not glycogenolysis reagents.

Educational objective:

Triglycerides are storage lipid molecules broken down (catabolized) by lipases into glycerol and three fatty acids. Glycerol is converted into dihydroxyacetone phosphate and incorporated into glycolysis or gluconeogenesis (anabolic pathway), depending on the cell's needs. The fatty acids diffuse into the mitochondria for β -oxidation, their major catabolic pathway.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403572
System:
Metabolic Reactions

Leptin signaling is vital for maintaining adequate body mass and preventing insulin resistance. Leptin acts on the sympathetic nervous system, which exerts long-lasting effects downstream on target tissues like muscle; however, it has been hypothesized that leptin may act on muscle tissue directly as well. When energy stores are abundant, adipocytes produce leptin to stimulate fatty acid oxidation and glucose utilization. One result of leptin signaling in skeletal muscle cells is the phosphorylation and activation of adenosine monophosphate–activated protein kinase (AMPK). AMPK facilitates the breakdown of triglycerides by decreasing the activity of acetyl-CoA carboxylase (ACC), which leads to downregulation of fatty acid synthesis and the upregulation of β -oxidation.

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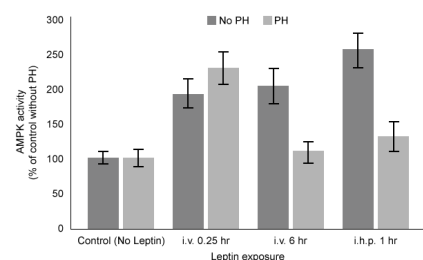


Figure 1 AMPK activity ($\pm$ standard error of the mean) measured in soleus muscle injected with leptin and PH via i.v. and i.h.p.

Table 1 ACC Activity in Soleus Muscle Exposed to Leptin Delivered via i.v. and i.h.p Injections *in vivo*

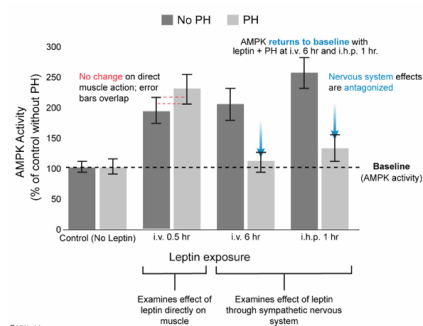
Injection	ACC activity (nmol g ⁻¹ min ⁻¹)	
	0 mM citrate	2 mM citrate
Saline 6 hr i.v.	12	45
Leptin 6 hr i.v.	8	22
Leptin 1 hr i.h.p.	7	18

What best describes the action of PH on AMPK activity?

- ☐ A. PH stimulates leptin action on the nervous system. [6%]
- ☐ B. PH antagonizes direct leptin action on muscles. [27%]
- ☒ C. PH antagonizes leptin action on the nervous system. [49%]
- ☐ D. PH stimulates direct leptin action on muscles. [17%]

Correct
49% answered correctly

Explanation:



According to the passage, **leptin** stimulates AMPK activity by activating the **sympathetic nervous system**, and it can also act directly on skeletal muscles. **Intrahypothalamic** (i.h.p.) injections allow leptin to be administered directly to the **central nervous system**, but **muscles are not directly affected** in this scenario. Intravenous (i.v.) injections administer leptin systemically—leptin travels through the blood to the muscles, but it also reaches the central nervous system. However, the passage also states that i.v. leptin is cleared quickly ($t_{1/2} = 40.2$ minutes).

Figure 1 shows the effects of i.v. versus i.h.p. leptin administration at various timepoints. Two timepoints are measured for i.v. administration: a short timepoint that corresponds to i.v. leptin binding leptin receptors on muscle cells, and a **longer timepoint**. By 6 hours (**~9 half-lives**), *all* leptin in the blood will have cleared, and any effects must correspond to long-lasting effects from sympathetic nervous system **hormones**.

Without PH, leptin stimulates AMPK activity at all timepoints measured for both routes. However, with PH, AMPK activity is kept at baseline in the single i.h.p. measurement, as well as 6 hours after i.v. administration. Because PH blocks the i.h.p. response, this must mean that **PH blocks** (ie, antagonizes) **the nervous system effects of leptin**. Similarly, **blocking of the long-term (i.v. 6 hr)**, but not short-term (i.v. 0.25 hr), response of i.v. leptin **confirms this effect**.

(Choice A) If PH stimulated leptin action on the nervous system, AMPK activity would be expected to *increase, not decrease*, following i.h.p. administration and 6 hours after i.v. administration.

(Choice B) If PH antagonized direct leptin action on muscles, the short-term i.v. effect at 0.25 hours with PH would be expected to *decrease* compared to the trial without PH.

(Choice D) If PH stimulated direct leptin action on muscles, AMPK activity would be expected to *significantly increase* with the short-term i.v. response at 0.25 hrs. Although a small increase is seen, the **error bars** corresponding to the data with and without PH overlap, indicating that this difference is *not statistically significant*, and that *no conclusion can be drawn*.

Educational objective:

Hormones (eg, leptin) are signaling molecules that can travel through the bloodstream, acting on tissues throughout the body to cause varied physiological effects (eg, increased fatty acid oxidation). Certain antagonists block the natural activity of hormones, and the route of administration of these antagonists (eg, intravenous vs. intrahypothalamic) can influence their effects.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403573
System:
Metabolic Reactions

Leptin signaling is vital for maintaining adequate body mass and preventing insulin resistance. Leptin acts on the sympathetic nervous system, which exerts long-lasting effects downstream on target tissues like muscle; however, it has been hypothesized that leptin may act on muscle tissue directly as well. When energy stores are abundant, adipocytes produce leptin to stimulate fatty acid oxidation and glucose utilization. One result of leptin signaling in skeletal muscle cells is the phosphorylation and activation of adenosine monophosphate-activated protein kinase (AMPK). AMPK facilitates the breakdown of triglycerides by decreasing the activity of acetyl-CoA carboxylase (ACC), which leads to downregulation of fatty acid synthesis and the upregulation of β -oxidation.

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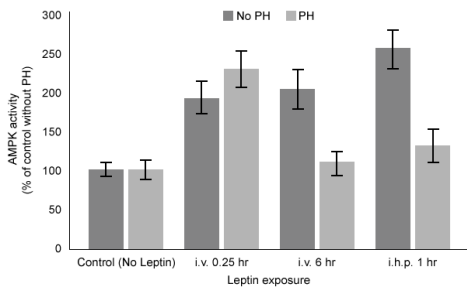


Figure 1 AMPK activity ($\pm$ standard error of the mean) measured in soleus muscle injected with leptin and PH via i.v. and i.h.p.

Table 1 ACC Activity in Soleus Muscle Exposed to Leptin Delivered via i.v. and i.h.p Injections *in vivo*

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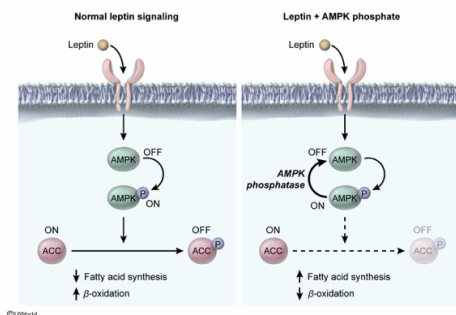
The addition of AMPK phosphatases to skeletal muscles exposed to leptin would result in which of the following?

- ☐ A. Increased shuttling of fatty acids from the cytosol into mitochondria [25%]
- ☐ B. Increased funneling of electrons to carriers NAD⁺ and FAD [9%]
- ☒ C. Upregulated activity of the acyl carrier protein in fatty acid synthase [38%]
- ☐ D. Upregulated phosphorylation of acetyl-coA carboxylase [26%]

Correct

38% answered correctly

Explanation:



Phosphatases remove phosphate groups from enzymes/substrates (ie, dephosphorylation) whereas kinases add phosphate groups (ie, phosphorylation). The passage states that phosphorylation activates AMPK; in this scenario, AMPK phosphatases in leptin-exposed muscles would *dephosphorylate* AMPK and thereby *inactivate* it. Inactive AMPK would then be **unable to phosphorylate and inhibit ACC (Choice D)**. Without inhibition, ACC will then carry out the first and rate-limiting step of **fatty acid synthesis** by carboxylating acetyl-CoA to malonyl-CoA, promoting fatty acid synthesis and preventing β -oxidation.

Fatty acid synthase carries out the main reactions of fatty acid synthesis and contains an **acyl carrier protein (ACP)** responsible for transferring acyl intermediates. Acetyl-CoA and malonyl-CoA are activated to acetyl-ACP and malonyl-ACP, which are condensed into acetoacetyl-ACP. Subsequently, acetoacetyl-ACP is reduced, dehydrated, and further reduced to generate butyryl-ACP. The coenzyme NADPH is used for both reductive steps. The reactions are repeated until a 16-carbon fatty acid (palmitate) is formed via addition of two carbon atoms per cycle. Because ACC activity is uninhibited in the presence of AMPK phosphatases, fatty acid synthesis is stimulated and **the activity of the acyl carrier protein is increased**.

(Choice A) In **β -oxidation**, a carnitine transporter moves fatty acids from the cytosol into the mitochondria. However, the presence of AMPK phosphatases would *inhibit* β -oxidation. Instead, fatty acid synthesis requires shuttling of citrate from the mitochondria into the cytosol to yield the acetyl-CoA precursor.

(Choice B) NAD^+ and FAD are reduced (ie, gain electrons) to become NADH and FADH_2 , respectively, in β -oxidation. However, fatty acid *synthesis* requires NADPH (ie, as an electron donor) to reduce the acyl groups added to the fatty acid.

Educational objective:

In the cytosol, ACC catalyzes the first and rate-limiting step of fatty acid synthesis by carboxylating acetyl-CoA to malonyl-CoA. Fatty acid synthase then carries out the main reactions of fatty acid synthesis. Fatty acid synthase contains an acyl carrier protein, which transfers acyl intermediates.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403574
System:
Metabolic Reactions

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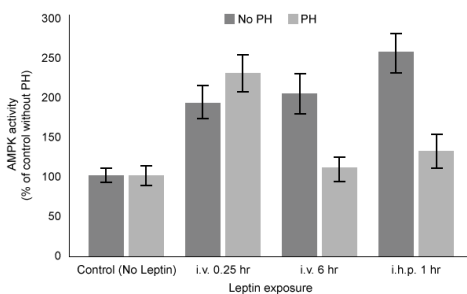


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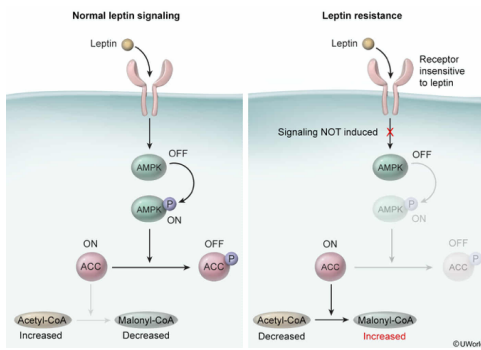
In skeletal muscle cells, leptin resistance would lead to decreased concentrations of:

- ☐ A. triglycerides. [12%]
- ☐ B. fatty acids. [22%]
- ☐ C. malonyl-CoA. [15%]
- ☒ D. acetyl-CoA. [49%]

Correct

50% answered correctly

Explanation:



Leptin resistance or leptin insensitivity refers to an inability to increase fatty acid oxidation in a cell despite adequate leptin production and normal binding of leptin to cell surface receptors. The passage states that leptin generally activates AMPK, which then phosphorylates/inhibits ACC.

If leptin were administered to skeletal muscles with leptin resistance, the exogenous leptin would fail to activate the AMPK signaling pathway. AMPK would therefore remain inactive and unable to block ACC activity. As a result, ACC would remain uninhibited and continue to carboxylate free acetyl-CoA molecules to produce malonyl-CoA, which would increase in concentration (**Choice C**). The levels of **acetyl-CoA would be reduced** due to its continued conversion to malonyl-CoA, resulting in increased fatty acid synthesis and decreased β -oxidation.

(Choices A and B) Leptin resistance leads to the accumulation of fatty acid precursors (eg, malonyl-CoA), resulting in an increased concentration of fatty acids, which then leads to an increased concentration of triglycerides. Furthermore, increased ACC activity as a result of leptin resistance also leads to reduced oxidation (ie, breakdown) of fatty acids.

Educational objective:

Leptin increases fatty acid oxidation and decreases fatty acid synthesis. In contrast, leptin-resistant cells cannot increase fatty acid oxidation despite adequate leptin signaling. As a result, AMPK signaling cannot be activated by leptin, which leads to increased and uninhibited ACC activity, promoting fatty acid synthesis and reducing β -oxidation.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403575
System:
Metabolic Reactions

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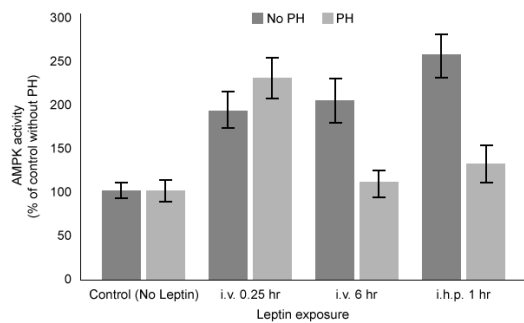


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What are the most likely role and effect of citrate on ACC activity?

- ☐ A. A cofactor that activates ACC [18%]
- ☐ B. A coenzyme that activates ACC [18%]

- ✔ ☒ C. An allosteric effector that activates ACC [47%]
☐ D. An inhibitor that inactivates ACC [15%]

Correct

48% answered correctly

Explanation:

Injection	ACC activity (nmol g ⁻¹ min ⁻¹)		Change in ACC activity due to 2 mM citrate (nmol g ⁻¹ min ⁻¹)
	0 mM citrate	2 mM citrate	
Saline 6 hr i.v.	12	45	↑ 33
Leptin 6 hr i.v.	8	22	↑ 14
Leptin 1 hr i.h.p.	7	18	↑ 11

Allosteric effectors regulate enzymatic activity by **binding** at sites **other than the active site**, causing **conformational changes** in the enzyme. As a result, enzymatic activity is either **decreased** (ie, inhibited) or **increased** (ie, activated). Table 1 shows that regardless of saline or leptin administration, ACC activity increases when citrate (2 mM) is added; thus, citrate must be an **activator of ACC**. Of the options shown, citrate is most likely an **allosteric effector** that **activates ACC**. Note that *any* increase in enzyme activity is considered activation, even if the enzyme had nonzero activity in the absence of the allosteric effector.

(Choices A and B) Cofactors, including metal ions and coenzymes (small organic compounds), are nonprotein compounds that are *required* for enzymatic activity. The ACC enzyme is active in the absence of citrate, so citrate is neither a coenzyme nor a cofactor.

(Choice D) Inhibitors *decrease* enzyme activity. Table 1 indicates that the addition of citrate (2 mM) corresponds to increased ACC activity.

Educational objective:

Allosteric effectors regulate enzymatic activity by binding at sites other than the active site, causing conformational changes in the enzyme. As a result, enzymatic activity is either inhibited (ie, decreased) or activated (ie, increased).

Time spent: 0 sec

Subject:
Biochemistry

QID: 403576

System:
Metabolic Reactions